

Cloud Computing Fundamentals: A Beginner's Guide

What is cloud computing?

Cloud computing means renting computing resources such as servers, storage, databases and software over the internet instead of buying and running your own hardware. You pay only for what you use, similar to paying for electricity. The three largest providers are Amazon Web Services (AWS), Microsoft Azure and Google Cloud, which together hold about 65% of the global cloud market.

Why companies move to the cloud

- Lower upfront cost: no need to buy servers; costs become a monthly operating expense.
- Scalability: add or remove capacity in minutes when traffic grows or drops.
- Reliability: providers run data centers in many regions, so a single failure does not stop the service.
- Speed: a new server can be ready in under 2 minutes instead of weeks for physical hardware.

The three service models

1. IaaS (Infrastructure as a Service): you rent virtual machines, networks and storage and manage the operating system yourself. Example: Amazon EC2.
2. PaaS (Platform as a Service): the provider manages servers and the operating system; you only deploy your code. Example: Google App Engine.
3. SaaS (Software as a Service): ready-to-use applications accessed through a browser. Examples: Gmail, Microsoft 365, Zoom.

Deployment models

- Public cloud: shared infrastructure owned by a provider, the most common and cheapest option.
- Private cloud: infrastructure dedicated to one organization, used when data rules are strict.
- Hybrid cloud: a mix of both, keeping sensitive data private while using the public cloud for everything else.

The shared responsibility model

Security in the cloud is shared. The provider secures the physical data centers, hardware and core network. The customer is responsible for their data, user accounts, passwords and access settings. Most cloud data breaches are caused by customer misconfiguration, such as a storage bucket left open to the public.

Best practices for using the cloud safely

1. Turn on multi-factor authentication for every account.
2. Give each person only the permissions they need (the principle of least privilege).
3. Encrypt sensitive data at rest and in transit.
4. Set budget alerts so unexpected usage does not create a surprise bill.
5. Back up important data to a second region.

Key takeaway

The cloud lets any team get world-class infrastructure on demand, but you stay responsible for configuring it securely and keeping costs under control.